Network Analysis and Economic Complexity for Sustainable Development

Final Assignment Report

Group 4

Section 1: Descriptive Statistics

The analysis of economic complexity begins with an examination of the raw data: the diversification of countries and the ubiquity of products.

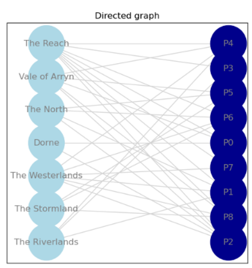
Following is the input data

```python
# Define matrix, connecting (imaginary) countries and activities.

countries=['Vale of Arryn','The North','The Riverlands', 'Dorne', 'The Stormland','The Westerlands','The Reach']
activities=[f'P{j}' for j in range(9)]

matrix = [[1, 0, 1, 0, 1, 1, 0, 0, 1],
          [0, 1, 1, 0, 0, 1, 1, 0, 0],
          [0, 1, 0, 1, 1, 0, 0, 0, 0],
          [1, 0, 1, 0, 0, 0, 0, 0, 0],
          [0, 0, 0, 0, 0, 1, 1, 0, 1],
          [0, 1, 1, 0, 1, 0, 1, 1, 1],
          [1, 1, 1, 1, 1, 0, 1, 1, 1]]
```

A bipartite graph framework is utilized to model the macroeconomic trade ecosystem, where countries (C = 7) and distinct product/activity classifications (P = 9) form two disjoint node sets. The relational linkages within this network are formalized via a binary adjacency matrix M. Specialization is defined by the condition $M_{cp} = 1$, which is satisfied when a country c achieves a Revealed Comparative Advantage index of $RCA \geq 1$for product p. Conversely, an entry $M_{cp} = 0$ signifies a lack of specialization.

The original network components, listing all participating countries and their corresponding activity/product classifications, are displayed in the Figure 1. This visualization clearly illustrates the interconnected relationship and direct mapping between each origin country and its designated activity nodes.

```
Original countries:
        index ' 0' - Vale of Arryn
        index ' 1' - The North
        index ' 2' - The Riverlands
        index ' 3' - Dorne
        index ' 4' - The Stormland
        index ' 5' - The Westerlands
        index ' 6' - The Reach
List of activities:
        index ' 0' - P0
        index ' 1' - P1
        index ' 2' - P2
        index ' 3' - P3
        index ' 4' - P4
        index ' 5' - P5
        index ' 6' - P6
        index ' 7' - P7
        index ' 8' - P8
```




```
Countries sorted by diversification (Most diverse at the top):
        1. - The Reach          (diversification: 8)
        2. - The Westerlands    (diversification: 6)
        3. - Vale of Arryn      (diversification: 5)
        4. - The North          (diversification: 4)
        5. - The Stormland      (diversification: 3)
        6. - The Riverlands     (diversification: 3)
        7. - Dorne              (diversification: 2)
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```

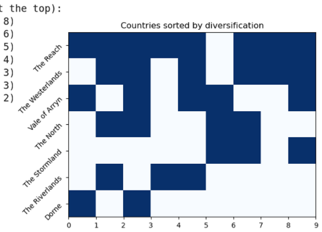


Figure 1. Original Countries and Activities/Products

Figure 2. Countries Sort by Diversification

Figure 2 illustrates a clear hierarchy in regional capability, where **The Reach** dominates with the highest diversification score of 8, followed closely by **The Westerlands** and **Vale of Arryn**. Furthermore, the heatmap displays a nested structural pattern (more distinctly in Figure 6 of next section) where less diversified regions, such as **Dorne**, limit their participation to a narrow subset of activities/products that are already heavily shared by the highly diversified economies at the top of the chart.

The network exhibits a clear nested, triangular structure driven by variations in product ubiquity (k_p). Highly ubiquitous commodities, such as **P2**, represent basic goods requiring standard capabilities that nearly all nations can produce. In contrast, rare products like **P3** and **P7** signify exclusive goods accessible only to regions with specialized capabilities. This structural nesting differentiates countries' advancement. Developing economies like **Dorne** remain restricted to common products, whereas diversified economies like **The Reach** manufacture both foundational commodities and exclusive, high-complexity products.

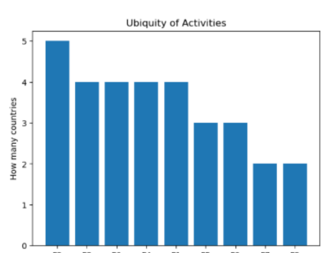


Figure 3. Ubiquity of products

Section 2: Economic Complexity Index (ECI) and Fitness

The Economic Complexity Index (ECI) was computed using the eigenvalue method to rank countries' economies by sophistication. **The Riverlands** achieves the highest ECI score (2.0548), indicating the highest level of structural complexity of its products, followed by **The Reach** (0.5441) and **The Westerlands** (0.3049). Conversely, **The Stormland** ranks lowest at -1.1651.

	Country	Diversification	ECI	Fitness	ECI Ranking	Fitness Ranking	Difference
1	The Riverlands	3	2.054840	0.973252	1	3	-2
2	The Reach	8	0.544078	2.268360	2	1	1
3	The Westerlands	6	0.304887	1.618939	3	2	1
4	The North	4	-0.481505	0.633110	4	5	-1
5	Vale of Arryn	5	-0.561359	0.784084	5	4	1
6	Dorne	2	-0.695802	0.209205	6	7	-1
7	The Stormland	3	-1.165138	0.513049	7	6	1

Figure 4. Comparison of ECI and Fitness

The economic complexity of a country is directly connected to the complexity of the products it exports. The Product Complexity Index (PCI) ranks the diversity and sophistication of the productive know-how required to manufacture an item. Based on the eigenvalue calculations, **P3** is the most complex product (1.961), followed by **P1** (0.7778) and **P4** (0.7438), which represent high-PCI, exclusive goods. Conversely, **P5** ranks lowest (-1.5098), representing a basic commodity.

Notably, the top three high-complexity products are all manufactured by **The Riverlands**. This concentration directly mirrors the country's first-place position in the ECI list, proving that a country's economic sophistication is structurally tethered to its output of high-PCI goods.

	Activity	Ubiquity	PCI	Complexity	PCI Ranking	Complexity Ranking	Difference
1	P3	2	1.961043	1.961588	1	2	-1
2	P1	4	0.777844	0.785781	2	4	-2
3	P4	4	0.743803	0.856881	3	3	0
4	P7	2	0.469049	2.721019	4	1	3
5	P2	5	-0.558192	0.331338	5	9	-4
6	P6	4	-0.594818	0.627909	6	6	0
7	P8	4	-0.628859	0.672498	7	5	2
8	P0	3	-0.660084	0.443413	8	8	0
9	P5	3	-1.509787	0.599571	9	7	2

Figure 5. Comparison of PCI and Complexity

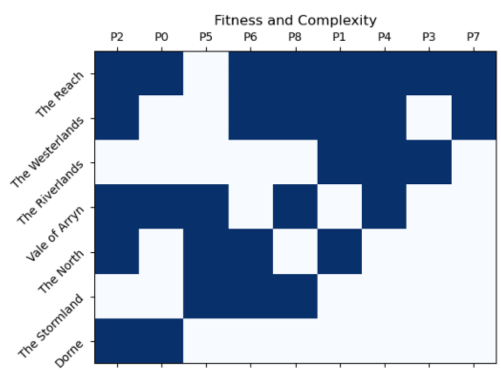


Figure 6. Fitness and Complexity

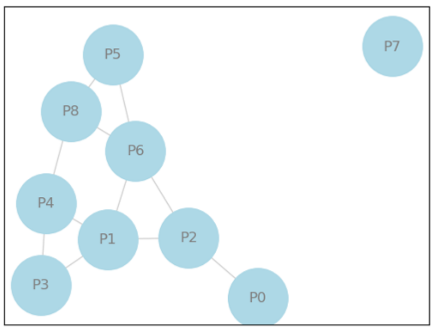
The structural comparison between standard Economic Complexity Index (ECI) calculations and non-linear Fitness metrics yields distinct country evaluations. According to standard ECI rankings, **The Riverlands** achieves the highest architectural complexity score, heavily supported by its highly specialized production of top-end goods like **P1**, **P4**, and **P3**.

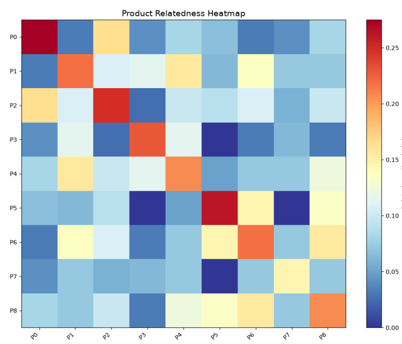
On the other hand, the Fitness and Complexity figure show that **The Reach** and **The Westerlands** emerge as the most developed countries due to their diverse, resilient industrial profiles and maximum activity breadth. In contrast, simpler countries like **Dorne** rely entirely on highly ubiquitous items (**P0, P2**), highlighting a fundamental structural gap in productive capabilities.

Section 3: Relatedness Between Activities

The relatedness between activities reveals the structural architecture of the product space, which is how capabilities are distributed and how countries can navigate toward more sophisticated production. This element will be analyzed in two complementary ways: the product space network (qualitative) and the relatedness heatmap (quantitative).

The product space network (Figure 7) illustrates the structural interconnectedness between the nine activities (P0-P8), revealing three distinct product roles. First, **P2** acts as the highly ubiquitous foundational baseline linked directly to **P0**. This positioning suggests that the capabilities required for **P2** are closely aligned with or prerequisite to **P0**, meaning every country capable of producing **P2** also possesses the capability to produce **P0**. Furthermore, **P1** and **P6** are situated as strategic entry points within the network. These nodes function as gateways connecting foundational items to a much wider, diverse range of industrial activities. On the other hand, **P7** stands completely isolated from the rest of the network. It registers as the least ubiquitous product, with only two countries having successfully developed the exclusive, highly specialized capabilities required to manufacture it.






```
Average Relatedness by Product:
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P0         : 0.091
P1         : 0.108
P2         : 0.111
P3         : 0.072
P4         : 0.108
P5         : 0.090
P6         : 0.108
P7         : 0.067
P8         : 0.108
```

Figure 7. Product Space Network

Figure 8: Product Relatedness Heatmap and Average Relatedness

The product relatedness heatmap (Figure 8) quantifies the structural relationships within the product space. Product **P2** exhibits the strongest relatedness (0.111), confirming its role as the foundational product that connects to the widest range, followed by **P1, P4, P6** and **P8** (0.108). In contrast, **P7** is the least related (0.067), being the most isolated product that requires unique and non-transferable capabilities to produce, along with **P3** (0.072) and **P5** (0.090).

Together, the analysis revealed a clear three-tier hierarchy. Developing nations, such as **Dorne** and **The Stormland**, occupy the bottom tier with low product diversification and inability to produce gateway products or complex products, while transitioning countries are positioned in the middle tier to build intermediate capabilities. Finally, advanced economies like **The Reach**, **The Westerlands** and **The Riverlands** populate the top tier, successfully securing capabilities to access the complex activities such as **P3** or **P7**. This hierarchy explains the ECI rankings: **The Riverlands** achieves the highest ECI precisely because it produces the two most isolated products (**P3** and **P7**).

Section 4: Diversification Strategies

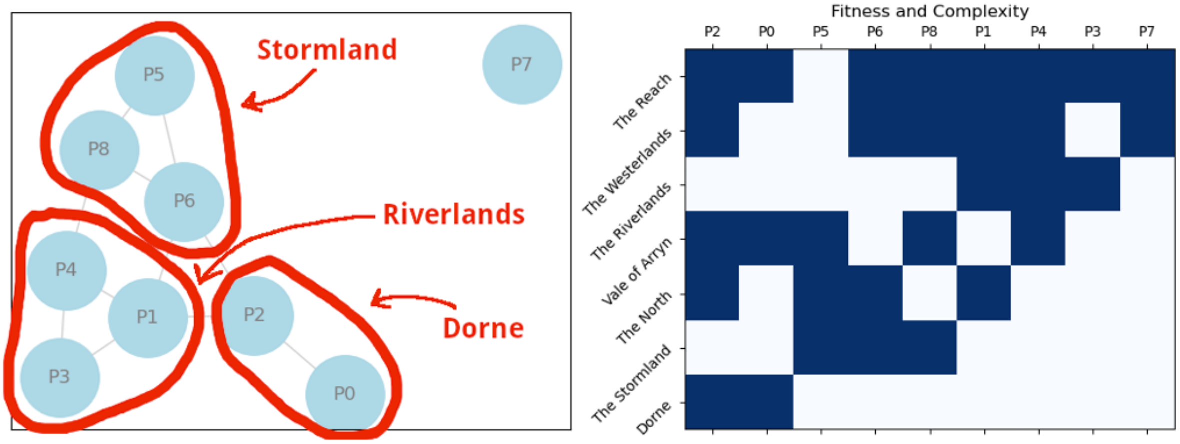

Figure 9. Detailed Mapping on Product Space Network

Tailored diversification paths are required based on each country's current positioning within the product space network to maximize economic complexity and fitness. For **Dorne**, which is currently restricted to the foundational core of highly ubiquitous products like **P0** and **P2**, the most viable strategic path is to target **P6**. Developing capabilities in **P6** functions as a critical bridge, allowing **Dorne** to escape its current limitations and opening doors to a wider array of interconnected industrial activities (albeit in the low-tier complexity of products).

For **The Stormland**, the best diversification path in the short term would be simple products **P2** and **P0**, whereas in the long term, transitioning into **P1** serves as a strategic gateway, granting the country access to less ubiquitous, highly sophisticated products located in the upper segments of the network map.

In contrast, **The Riverlands** already specializes in top-tier complex products such as **P1**, **P3**, and **P4**. Its optimal strategy focuses on resilience and broad-based market stability rather than upward leaps. **The Riverlands** should target **P6**, which will allow it to successfully bridge into and access the more ubiquitous, high-volume products of the lower segments. This dual-presence strategy enhances economic stability by anchoring its sophisticated industrial core to foundational trade activities.

In conclusion, successful economic diversification relies entirely on leveraging product-relatedness within the network. By adopting these targeted, capability-driven strategies, each country can effectively bridge structural gaps, enhance industrial resilience, and maximize long-term economic growth.

Section 5: Workflow & Contributions

Team Organization

Our team (Group 4) organized our work through a series of conversations between July 3 and July 10. Initial discussions started via email and then changed to Google Chat for asynchronous communication.

Decision-Making

Key decisions were made collectively:

- **Data:** We used the data provided in the assignment.
- **Methods:** We implemented both Method of Reflections and eigenvector ECI, PCI to demonstrate methodological differences and handle sign indeterminacy. Fitness and Complexity methods are used to demonstrate relatedness and nestedness and to compare with ECI and PCI values in terms of ranking.
- **Visualization:** We prioritized clear, readable graphics with appropriate labels.
- **Report Structure:** We followed the assignment's five-section structure with one page per section.

Individual Contributions

Team Member	Contribution
Petr Ospaly	Organizing, making python notebook, initial slides and interpretation
Truong Ngan Giang	Data visualization, interpretation and report writing
Mesay Unitira	Data interpretation and report writing
Carl Billdjel Bahian	Performed computational analyses and interpretation
Pyaekyita	Assisted in the preparation and development of the report

All team members participated in reviewing the final report and presentation.